



1 Nonproprietary Names

BP: Acacia

JP: Acacia

PhEur: Acacia

USP-NF: Acacia

2 Synonyms

Acaciae gummi; acacia gum; arabic gum; E414; gum acacia; gummi africanum; gum arabic; gummi arabicum; gummi mimosae; talha gum.

3 Chemical Name and CAS Registry Number

Acacia [9000-01-5]

4 Empirical Formula and Molecular Weight

Acacia is a complex, loose aggregate of sugars and hemicelluloses with a molecular weight of approximately 240 000–580 000. The aggregate consists essentially of an arabic acid nucleus to which are connected calcium, magnesium, and potassium along with the sugars arabinose, galactose, and rhamnose.

5 Structural Formula

See Section 4.

6 Functional Category

Emulsifying agent; stabilizing agent; suspending agent; tablet binder; viscosity-increasing agent.

7 Applications in Pharmaceutical Formulation or Technology

Acacia is mainly used in oral and topical pharmaceutical formulations as a suspending and emulsifying agent, often in combination with tragacanth. It is also used in the preparation of pastilles and lozenges, and as a tablet binder, although if used incautiously it can produce tablets with a prolonged disintegration time. Acacia has also been evaluated as a bioadhesive;⁽¹⁾ and has been used in novel tablet formulations,⁽²⁾ and modified release tablets.⁽³⁾ See Table I.

Acacia is also used in cosmetics, confectionery, food products, and spray-dried flavors.⁽⁴⁾

See also Section 18.

Table I: Uses of acacia.

Use	Concentration (%)
Emulsifying agent	10–20
Pastille base	10–30
Suspending agent	5–10
Tablet binder	1–5

8 Description

Acacia is available as white or yellowish-white thin flakes, spheroidal tears, granules, powder, or spray-dried powder. It is odorless and has a bland taste.

9 Pharmacopeial Specifications

The PhEur 6.3 provides monographs on acacia and spray-dried acacia, while the USP32–NF27 describes acacia in a single monograph that encompasses tears, flakes, granules, powder, and spray-dried powder. The USP32–NF27 also has a monograph on acacia syrup. The JP XV has monographs on acacia and powdered acacia. See Table II.

Table II: Pharmacopeial specifications for acacia.

Test	JP XV	PhEur 6.3	USP32–NF27
Identification	+	+	+
Characters	+	+	+
Microbial limit	—	≤ 10 ⁴ cfu/g	+
Water	≤ 17.0% ≤ 15.0% ^(a)	≤ 15.0% ≤ 10.0% ^(b)	≤ 15.0% —
Total ash	≤ 4.0%	≤ 4.0%	≤ 4.0%
Acid-insoluble ash	≤ 0.5%	—	≤ 0.5%
Insoluble residue	≤ 0.2%	≤ 0.5%	≤ 50 mg/5g
Arsenic	—	—	≤ 3 ppm
Lead	—	—	≤ 0.001%
Heavy metals	—	—	≤ 0.004%
Starch, dextrin, and agar	—	+	+
Tannin-bearing gums	+	+	+
Tragacanth	—	+	—
Sterculia gum	—	+	—
Glucose and fructose	+	+	—
Solubility and reaction	—	—	+

(a) Powdered acacia.

(b) Spray-dried acacia.

10 Typical Properties

Acidity/alkalinity pH = 4.5–5.0 (5% w/v aqueous solution)

Acid value 2.5

Hygroscopicity At relative humidities of 25–65%, the equilibrium moisture content of powdered acacia at 25°C is 8–13% w/w, but at relative humidities above about 70% it absorbs substantial amounts of water.

NIR spectra see Figure 1.

Solubility Soluble 1 in 20 of glycerin, 1 in 20 of propylene glycol, 1 in 2.7 of water; practically insoluble in ethanol (95%). In water, acacia dissolves very slowly, although almost completely after two hours, in twice the mass of water leaving only a very small residue of powder. The solution is colorless or yellowish, viscous, adhesive, and translucent. Spray-dried acacia dissolves more rapidly, in about 20 minutes.

Specific gravity 1.35–1.49

Viscosity (dynamic) 100 mPa s (100 cP) for a 30% w/v aqueous solution at 20°C. The viscosity of aqueous acacia solutions varies depending upon the source of the material, processing, storage conditions, pH, and the presence of salts. Viscosity increases slowly up to about 25% w/v concentration and exhibits Newtonian behavior. Above this concentration, viscosity increases rapidly (non-Newtonian rheology). Increasing temperature or prolonged heating of solutions results in a decrease of viscosity owing to depolymerization or particle agglomeration. See also Section 12.

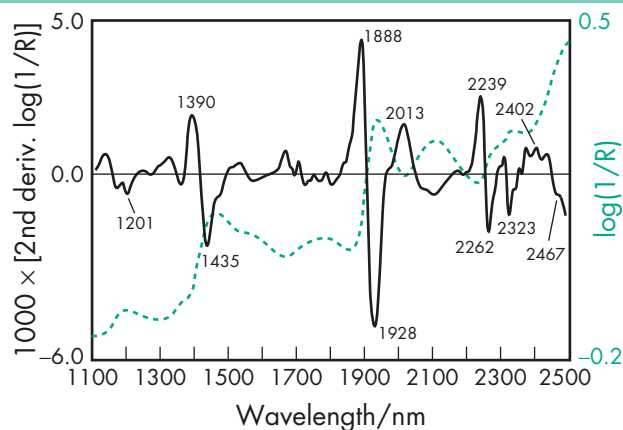


Figure 1: Near-infrared spectrum of acacia measured by reflectance.

11 Stability and Storage Conditions

Aqueous solutions are subject to bacterial or enzymatic degradation but may be preserved by initially boiling the solution for a short time to inactivate any enzymes present; microwave irradiation can also be used.⁽⁵⁾ Aqueous solutions may also be preserved by the addition of an antimicrobial preservative such as 0.1% w/v benzoic acid, 0.1% w/v sodium benzoate, or a mixture of 0.17% w/v methylparaben and 0.03% propylparaben. Powdered acacia should be stored in an airtight container in a cool, dry place.

12 Incompatibilities

Acacia is incompatible with a number of substances including amidopyrine, apomorphine, cresol, ethanol (95%), ferric salts, morphine, phenol, physostigmine, tannins, thymol, and vanillin.

An oxidizing enzyme present in acacia may affect preparations containing easily oxidizable substances. However, the enzyme may be inactivated by heating at 100°C for a short time; see Section 11.

Many salts reduce the viscosity of aqueous acacia solutions, while trivalent salts may initiate coagulation. Aqueous solutions carry a negative charge and will form coacervates with gelatin and other substances. In the preparation of emulsions, solutions of acacia are incompatible with soaps.

13 Method of Manufacture

Acacia is the dried gummy exudate obtained from the stems and branches of *Acacia senegal* (Linné) Willdenow or other related species of *Acacia* (Fam. Leguminosae) that grow mainly in the Sudan and Senegal regions of Africa.

The bark of the tree is incised and the exudate allowed to dry on the bark. The dried exudate is then collected, processed to remove bark, sand, and other particulate matter, and graded. Various acacia grades differing in particle size and other physical properties are thus obtained. A spray-dried powder is also commercially available.

14 Safety

Acacia is used in cosmetics, foods, and oral and topical pharmaceutical formulations. Although it is generally regarded as an essentially nontoxic material, there have been a limited number of reports of hypersensitivity to acacia after inhalation or ingestion.^(6,7) Severe anaphylactic reactions have occurred following the parenteral administration of acacia and it is now no longer used for this purpose.⁽⁶⁾

The WHO has not set an acceptable daily intake for acacia as a food additive because the levels necessary to achieve a desired effect were not considered to represent a hazard to health.⁽⁸⁾

LD₅₀ (hamster, oral): >18 g/kg⁽⁹⁾

LD₅₀ (mouse, oral): >16 g/kg

LD₅₀ (rabbit, oral): 8.0 g/kg

LD₅₀ (rat, oral): >16 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Acacia can be irritant to the eyes and skin and upon inhalation. Gloves, eye protection, and a dust respirator are recommended.

16 Regulatory Status

GRAS listed. Accepted for use in Europe as a food additive. Included in the FDA Inactive Ingredients Database (oral preparations and buccal or sublingual tablets). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

17 Related Substances

Ceratonia; guar gum; tragacanth.

18 Comments

Concentrated aqueous solutions are used to prepare pastilles since on drying they form solid rubbery or glasslike masses depending upon the concentration used. Foreign policy changes and politically unstable conditions in Sudan, which is the principal supplier of acacia, has created a need to find a suitable replacement.⁽¹⁰⁾ Poloxamer 188 (12–15% w/w) can be used to make an oil/water emulsion with similar rheological characteristics to acacia. Other natural by-products of foods can also be used.⁽¹¹⁾ Acacia is also used in the food industry as an emulsifier, stabilizer, and thickener. A specification for acacia is contained in the Food Chemicals Codex (FCC).⁽¹²⁾

The EINECS number for acacia is 232-519-5.

19 Specific References

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20 General References

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21 Author

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22 Date of Revision

10 February 2009.



Acesulfame Potassium

1 Nonproprietary Names

BP: Acesulfame Potassium
 PhEur: Acesulfame Potassium
 USP-NF: Acesulfame Potassium

2 Synonyms

Acesulfame K; ace K; acesulfamum kalicum; E950; 6-methyl-3,4-dihydro-1,2,3-oxathiazin-4(3H)-one-2,2-dioxide potassium salt; potassium 6-methyl-2,2-dioxo-oxathiazin-4-olate; *Sunett*; *Sweet One*.

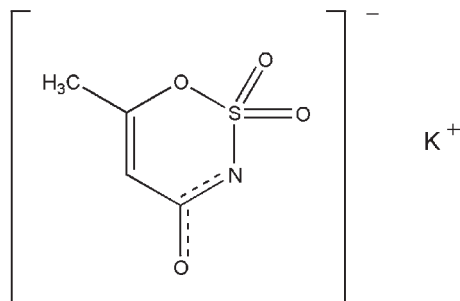
3 Chemical Name and CAS Registry Number

6-Methyl-1,2,3-oxathiazin-4(3H)-one-2,2-dioxide potassium salt [55589-62-3]

4 Empirical Formula and Molecular Weight

C₄H₄KNO₄S 201.24

5 Structural Formula



6 Functional Category

Sweetening agent.

7 Applications in Pharmaceutical Formulation or Technology

Acesulfame potassium is used as an intense sweetening agent in cosmetics, foods, beverage products, table-top sweeteners, vitamin and pharmaceutical preparations, including powder mixes, tablets, and liquid products. It is widely used as a sugar substitute in compounded formulations,⁽¹⁾ and as a toothpaste sweetener.⁽²⁾

The approximate sweetening power is 180–200 times that of sucrose, similar to aspartame, about one-third as sweet as sucralose, one-half as sweet as sodium saccharin, and about 4–5 times sweeter

than sodium cyclamate.⁽³⁾ It enhances flavor systems and can be used to mask some unpleasant taste characteristics.

8 Description

Acesulfame potassium occurs as a colorless to white-colored, odorless, crystalline powder with an intensely sweet taste.

9 Pharmacopeial Specifications

See Table I.

10 Typical Properties

Acidity/alkalinity pH = 5.5–7.5 (1% w/v aqueous solution)
Bonding index 0.007⁽⁴⁾
Brittle fracture index 0.08⁽⁴⁾
Density (bulk) 1.04 g/cm³ ⁽⁴⁾
Density (tapped) 1.28 g/cm³ ⁽⁴⁾
Elastic modulus 4000 MPa⁽⁴⁾
Flowability 19% (Carr compressibility index)⁽⁴⁾
Melting point 250°C
NIR spectra see Figure 1.
Solubility see Table II.

SEM 1: Excipient: acesulfame potassium; magnification: 150×; voltage: 5 kV.

